Unit: 6 Unit 6: Systems of Linear Equations and Inequalities **Name:** ________________
Topic: Solving Systems Using Elimination and Substitution **Date:** ____________

Homework (Jalapeno)

Q1. In ancient Egypt, $2x + 3y = 12$ and $x - 2y = -3$ represent the cost of building two types of temples. What method can be used to solve for x and y?
(A) Only substitution
(B) Only elimination
(C) Both substitution and elimination
(D) Neither substitution nor elimination
(E) Graphing

Q2. The ancient Greeks used $x + 4y = 10$ and $3x - 2y = 5$ to plan the construction of a theater. What is the value of y when solved using substitution?
(A) $y = 1$
(B) $y = 2$
(C) $y = 3$
(D) $y = 4$
(E) $y = 5$

Q3. The Rosetta Stone has an inscription with $2x - 5y = 11$ and $x + 2y = 5$. How many solutions does this system have?
(A) One solution
(B) No solution
(C) Infinitely many solutions
(D) Two solutions
(E) Three solutions

Q4. The ancient Chinese used $3x + 2y = 12$ and $x - 3y = -4$ to design a new type of pagoda. What is the value of x when solved using elimination?
(A) $x = 2$
(B) $x = 3$
(C) $x = 4$
(D) $x = 5$
(E) $x = 6$

Q5. The ancient Mayans used $x + 2y = 7$ and $4x - 3y = 10$ to plan a new trade route. What method is more efficient to solve this system?
(A) Substitution
(B) Elimination
(C) Graphing
(D) Both substitution and elimination
(E) Neither substitution nor elimination

Q6. The Terracotta Army was built using $2x - 3y = 8$ and $x + 4y = 12$. How many solutions does this system have?
(A) One solution
(B) No solution
(C) Infinitely many solutions
(D) Two solutions
(E) Three solutions

Q7. The Great Pyramid of Giza has a system of equations $3x + 2y = 15$ and $2x - 5y = -2$. What is the value of x when solved using substitution?
(A) $x = 2$
(B) $x = 3$
(C) $x = 4$
(D) $x = 5$
(E) $x = 6$

Q8. The Hanging Gardens of Babylon used $x - 2y = 3$ and $4x + 3y = 14$ to design a new irrigation system. What method can be used to solve for x and y?
(A) Only substitution
(B) Only elimination
(C) Both substitution and elimination
(D) Neither substitution nor elimination
(E) Graphing

Open Ended Questions (FRQ)

Q9. The ancient city of Pompeii has a system of equations $2x + 5y = 20$ and $3x - 2y = 5$. Solve for x and y using elimination and show your work.

Q10. The Colosseum in Rome has a system of equations $x + 3y = 12$ and $4x - 2y = 8$. Solve for x and y using substitution and show your work.

Chef's Tips

- Check for units and dimensions in the answer
- Ensure all work is shown for free response questions
- Use a calculator only when necessary